GROWMORE GROUP OF INSTITUTE

UNIT – 1

Operating System

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Introduction to operating System:

An **Operating System (OS)** is a software that acts as an interface between computer hardware components and the user. Every computer system must have at least one operating system to run other programs.

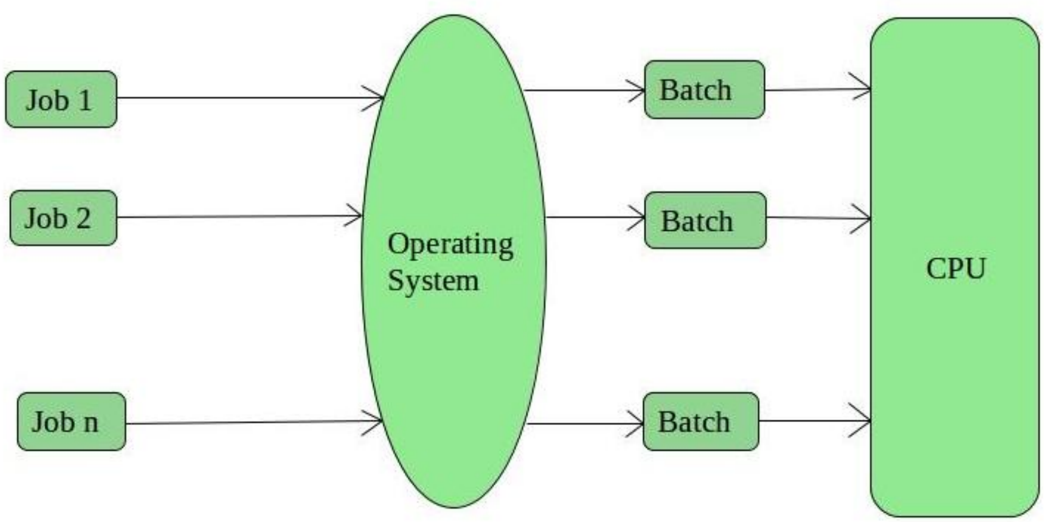
An Operating System (OS) is an interface between a computer user and computer hardware. An operating system is a software which performs all the basic tasks like file management, memory management, process management, handling input and output, and controlling peripheral devices such as disk drives and printers.

Types of operating system:

An Operating System performs all the basic tasks like managing files, processes, and memory. Thus operating system acts as the manager of all the resources, i.e. **resource manager**.

1. Batch Operating System –

This type of operating system does not interact with the computer directly. There is an operator which takes similar jobs having the same requirement and group them into batches. It is the responsibility of the operator to sort jobs with similar needs.



Advantages of Batch Operating System:

- Multiple users can share the batch systems
- The idle time for the batch system is very less
- It is easy to manage large work repeatedly in batch systems

Disadvantages of Batch Operating System:

- The computer operators should be well known with batch systems
- Batch systems are hard to debug
- It is sometimes costly
- The other jobs will have to wait for an unknown time if any job fails

2. Time-Sharing Operating Systems –

Each task is given some time to execute so that all the tasks work smoothly. Each user gets the time of CPU as they use a single system. These systems are also known as Multitasking Systems.

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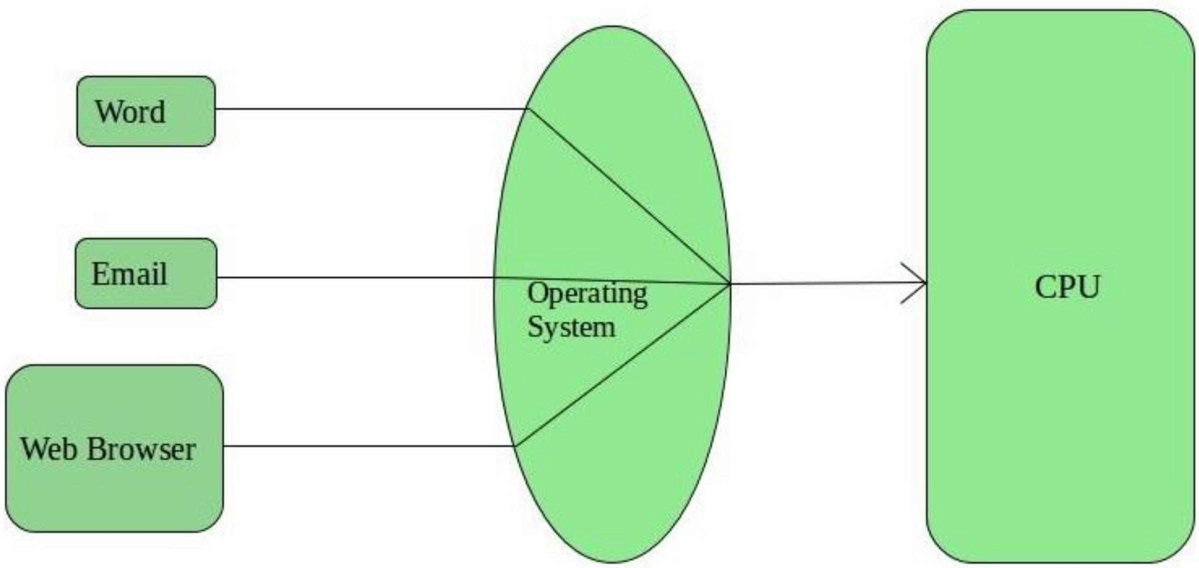


Advantages of Time-Sharing OS:

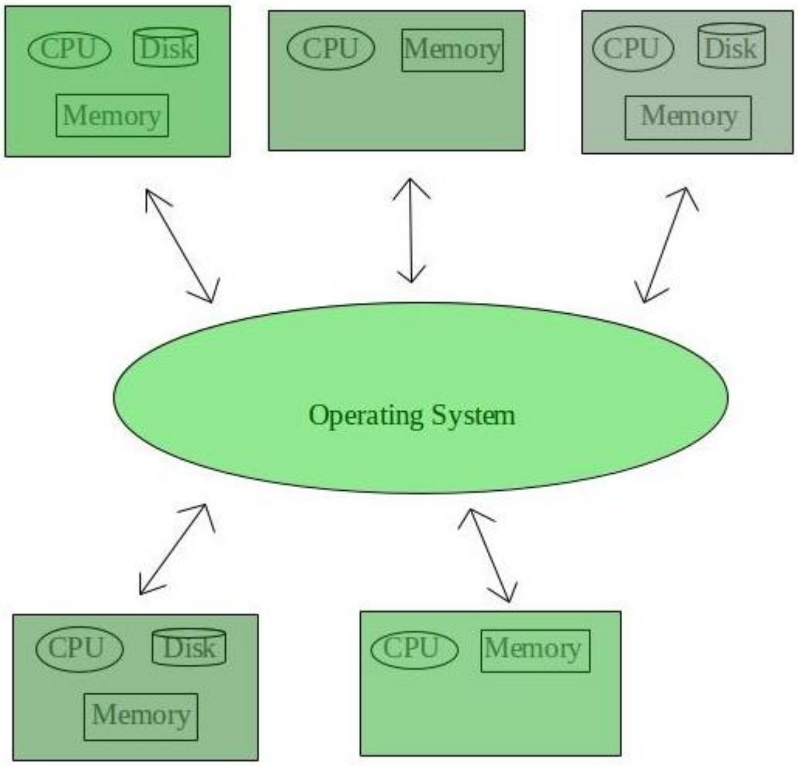
- Each task gets an equal opportunity
- Fewer chances of duplication of software
- CPU idle time can be reduced

Disadvantages of Time-Sharing OS:

- Reliability problem
- One must have to take care of the security and integrity of user programs and data
- Data communication problem

3. Distributed Operating System –

These types of the operating system is a recent advancement in the world of computer technology and are being widely accepted all over the world and, that too, with a great pace. Various autonomous interconnected computers communicate with each other using a shared communication network. Independent systems possess their own memory unit and CPU.



Advantages of Distributed Operating System:

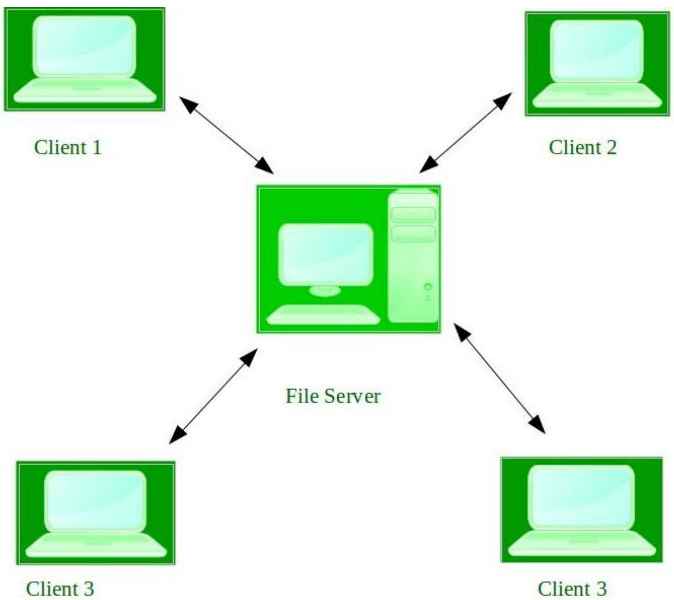
- Failure of one will not affect the other network communication, as all systems are independent from each other
- Electronic mail increases the data exchange speed
- Delay in data processing reduces

Disadvantages of Distributed Operating System:

- Failure of the main network will stop the entire communication
- To establish distributed systems the language which is used are not well defined yet
- These types of systems are not readily available as they are very expensive. Not only that the underlying software is highly complex and not understood well yet.

4. Network Operating System –

These systems run on a server and provide the capability to manage data, users, groups, security, applications, and other networking functions. These types of operating systems allow shared access of files, printers, security, applications, and other networking functions over a small private network.



Advantages of Network Operating System:

- Highly stable centralized servers
- Security concerns are handled through servers
- New technologies and hardware up-gradation are easily integrated into the system
- Server access is possible remotely from different locations and types of systems

Disadvantages of Network Operating System:

- Servers are costly
- User has to depend on a central location for most operations
- Maintenance and updates are required regularly

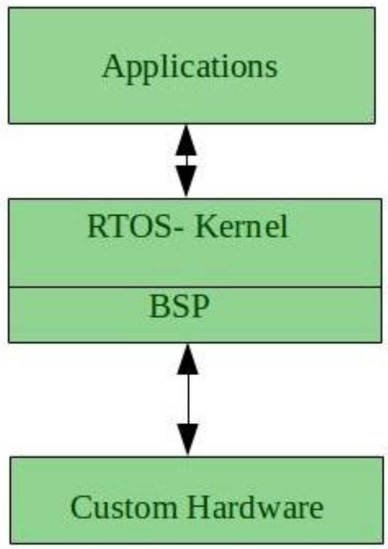
5. Real-Time Operating System –

These types of OSs serve real-time systems. The time interval required to process and respond to inputs is very small. This time interval is called **response time**.
Real-time systems are used when there are time requirements that are very strict like missile systems, air traffic control systems, robots, etc.

Two types of Real-Time Operating System which are as follows:

- **Hard Real-Time Systems:**
  These OSs are meant for applications where time constraints are very strict and even the shortest possible delay is not acceptable.

- **Soft Real-Time Systems:**
  These OSs are for applications where for time-constraint is less strict.



Advantages of RTOS:

- Maximum ConsumptionTask ShiftingFocus on ApplicationReal-time operating system in the embedded system:
- Error Free:
- Memory Allocation

Disadvantages of RTOS:

- Limited Tasks
- Use heavy system resources
- Complex Algorithms
- Device driver and interrupt signals
- Thread Priority

Components of operating system:

An operating system is a large and complex system that can only be created by partitioning into small parts. These pieces should be a well-defined part of the system, carefully defining inputs, outputs, and functions.

Although Windows, Mac, UNIX, Linux, and other OS do not have the same structure, most operating systems share similar OS system components, such as file, memory, process, I/O device management.

1. Process Management
2. File Management
3. Network Management
4. Main Memory Management
5. Secondary Storage Management
6. I/O Device Management
7. Security Management
8. Command Interpreter System

1.process management

The process management component is a procedure for managing many processes running simultaneously on the operating system. Every running software application program has one or more processes associated with them.

For example, when you use a search engine like Chrome, there is a process running for that browser program.

Process management keeps processes running efficiently. It also uses memory allocated to them and shutting them down when needed.

2.File management

A file is a set of related information defined by its creator. It commonly represents programs (both source and object forms) and data. Data files can be alphabetic, numeric, or alphanumeric.

File and directory creation and deletion.

For manipulating files and directories.

Mapping files onto secondary storage.

Backup files on stable storage media.

3.Network management

Network management is the process of administering and managing computer networks. It includes performance management, provisioning of networks, fault analysis, and maintaining the quality of service.

A distributed system is a collection of computers or processors that never share their memory and clock. In this type of system, all the processors have their local memory, and the processors communicate with each other using different communication cables, such as fibre optics or telephone lines.

4.Main memory management

Main memory is a large array of storage or bytes, which has an address. The memory management process is conducted by using a sequence of reads or writes of specific memory addresses.

It should be mapped to absolute addresses and loaded inside the memory to execute a program. The selection of a memory management method depends on several factors.

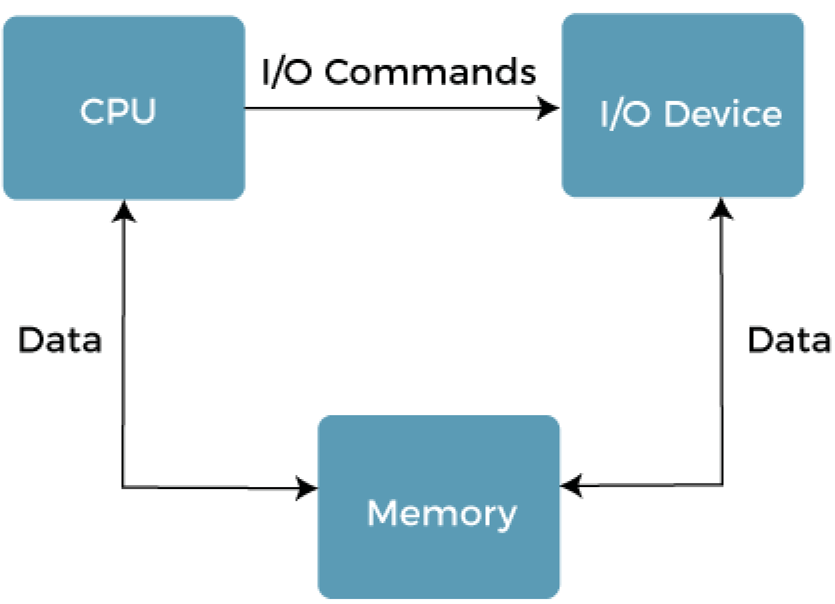
5. Secondary storage management

The most important task of a computer system is to execute programs. These programs help you to access the data from the main memory during execution. This memory of the computer is very small to store all data and programs permanently. The computer system offers secondary storage to back up the main memory.

Today modern computers use hard drives/SSD as the primary storage of both programs and data. However, the secondary storage management also works with storage devices, such as USB flash drives and CD/DVD drives.

6. I/O Device management

One of the important use of an operating system that helps to hide the variations of specific hardware devices from the user.



The I/O management system offers the following functions, such as:

- It offers a buffer caching system
- It provides general device driver code
- It provides drivers for particular hardware devices.
- I/O helps you to know the individualities of a specific device.

7. Security management

The various processes in an operating system need to be secured from other activities. Therefore, various mechanisms can ensure those processes that want to operate files, memory CPU, and other hardware resources should have proper authorization from the operating system.

Security refers to a mechanism for controlling the access of programs, processes, or users to the resources defined by computer controls to be imposed, together with some means of enforcement.

8.Command interpreter System

One of the most important components of an operating system is its command interpreter. The command interpreter is the primary interface between the user and the rest of the system.

- The control card interpreter,
- The command-line interpreter,
- The shell (in UNIX), and so on.

Services of operating System:

An Operating System provides services to both the users and to the programs.

- It provides programs an environment to execute.
- It provides users the services to execute the programs in a convenient manner.

Following are a few common services provided by an operating system −

- Program execution
- I/O operations
- File System manipulation
- Communication
- Error Detection
- Resource Allocation
- Protection

Program execution

Operating systems handle many kinds of activities from user programs to system programs like printer spooler, name servers, file server, etc. Each of these activities is encapsulated as a process.

- Loads a program into memory.
- Executes the program.
- Handles program's execution.
- Provides a mechanism for process synchronization.
- Provides a mechanism for process communication.
- Provides a mechanism for deadlock handling.

I/O Operation

An I/O subsystem comprises of I/O devices and their corresponding driver software. Drivers hide the peculiarities of specific hardware devices from the users.

An Operating System manages the communication between user and device drivers.

- I/O operation means read or write operation with any file or any specific I/O device.
- Operating system provides the access to the required I/O device when required.

File system manipulation

A file represents a collection of related information. Computers can store files on the disk (secondary storage), for long-term storage purpose. Examples of storage media include magnetic tape, magnetic disk and optical disk drives like CD, DVD. Each of these media has its own properties like speed, capacity, data transfer rate and data access methods.

- Program needs to read a file or write a file.
- The operating system gives the permission to the program for operation on file.
- Permission varies from read-only, read-write, denied and so on.
- Operating System provides an interface to the user to create/delete files.

- Operating System provides an interface to the user to create/delete directories.
- Operating System provides an interface to create the backup of file system.

Communication

In case of distributed systems which are a collection of processors that do not share memory, peripheral devices, or a clock, the operating system manages communications between all the processes. Multiple processes communicate with one another through communication lines in the network.

- Two processes often require data to be transferred between them
- Both the processes can be on one computer or on different computers, but are connected through a computer network.

Error handling

Errors can occur anytime and anywhere. An error may occur in CPU, in I/O devices or in the memory hardware. Following are the major activities of an operating system with respect to error handling −

- The OS constantly checks for possible errors.
- The OS takes an appropriate action to ensure correct and consistent computing.

Resource Management

In case of multi-user or multi-tasking environment, resources such as main memory, CPU cycles and files storage are to be allocated to each user or job. Following are the major activities of an operating system with respect to resource management −

- The OS manages all kinds of resources using schedulers.
- CPU scheduling algorithms are used for better utilization of CPU.

Protection

Considering a computer system having multiple users and concurrent execution of multiple processes, the various processes must be protected from each other's activities.

Protection refers to a mechanism or a way to control the access of programs, processes, or users to the resources defined by a computer system. Following are the major activities of an operating system with respect to protection −

- The OS ensures that all access to system resources is controlled.
- The OS ensures that external I/O devices are protected from invalid access attempts.
- The OS provides authentication features for each user by means of passwords.

Buffering in Operating System

The ***buffer*** is an area in the **main memory** used to store or hold the data **temporarily**. In other words, buffer temporarily stores data transmitted from one place to another, either between two devices or an application. The act of storing data temporarily in the buffer is called ***buffering***.

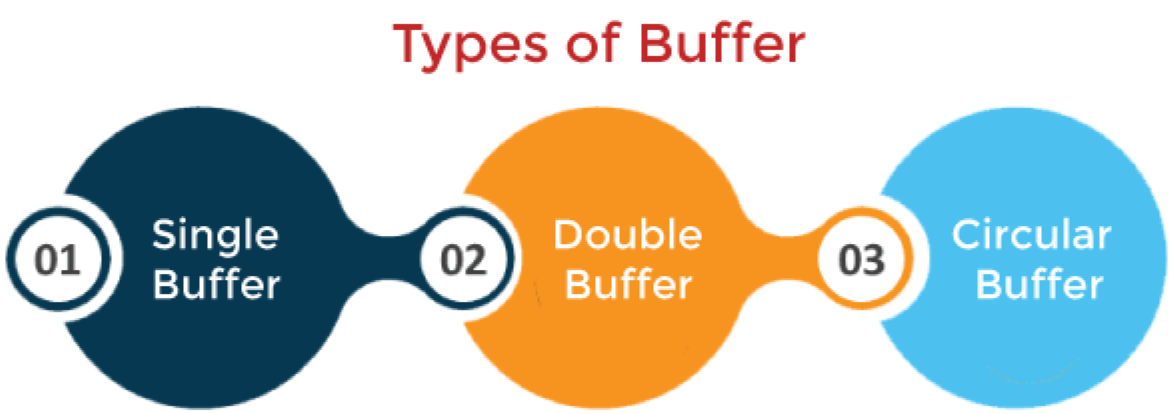
A buffer may be used when moving data between processes within a computer. Buffers can be implemented in a fixed memory location in hardware or by using a virtual data buffer in software, pointing at a location in the physical memory. In all cases, the data in a data buffer are stored on a physical storage medium.

Purpose of Buffering

You face buffer during watching videos on YouTube or live streams. In a video stream, a buffer represents the amount of data required to be downloaded before the video can play to the viewer in real-time. A buffer in a computer environment means that a set amount of data will be stored to preload the required data before it gets used by the CPU.

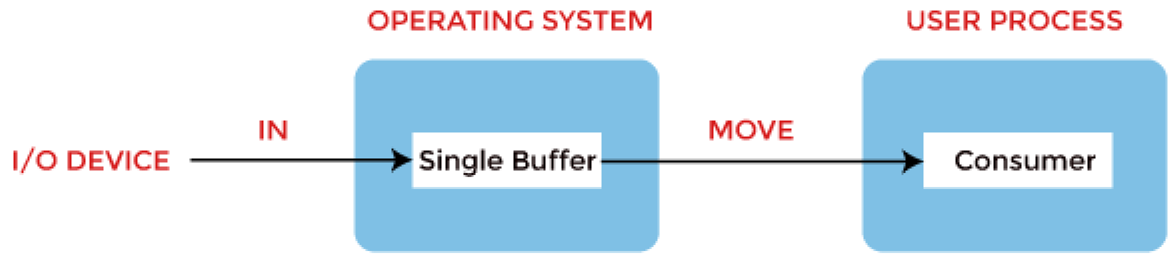
Types of Buffering

There are three main types of buffering in the operating system, such as:



1. Single Buffer

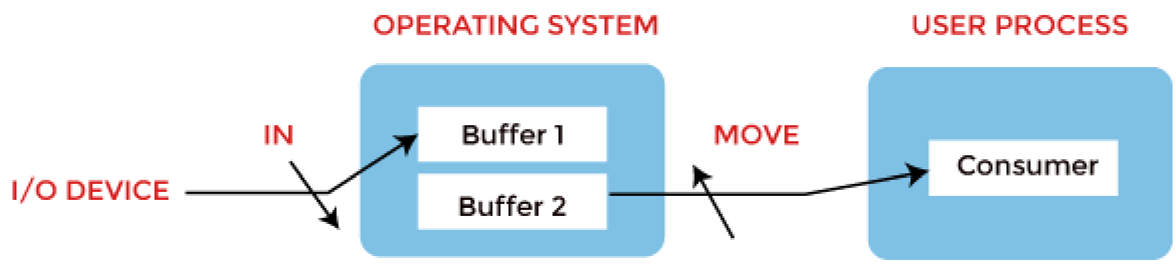
In Single Buffering, only one buffer is used to transfer the data between two devices. The producer produces one block of data into the buffer. After that, the consumer consumes the buffer. Only when the buffer is empty, the processor again produces the data.



SINGLE I/O BUFFERING

2. Double Buffer

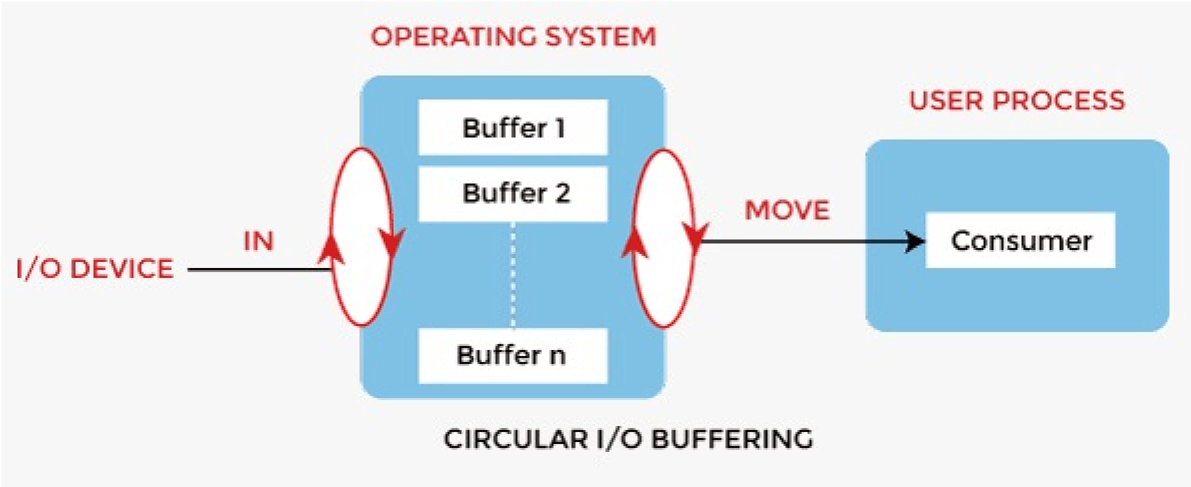
In **Double Buffering**, two schemes or two buffers are used in the place of one. In this buffering, the producer produces one buffer while the consumer consumes another buffer simultaneously. So, the producer not needs to wait for filling the buffer. Double buffering is also known as buffer swapping.



DOUBLE I/O BUFFERING

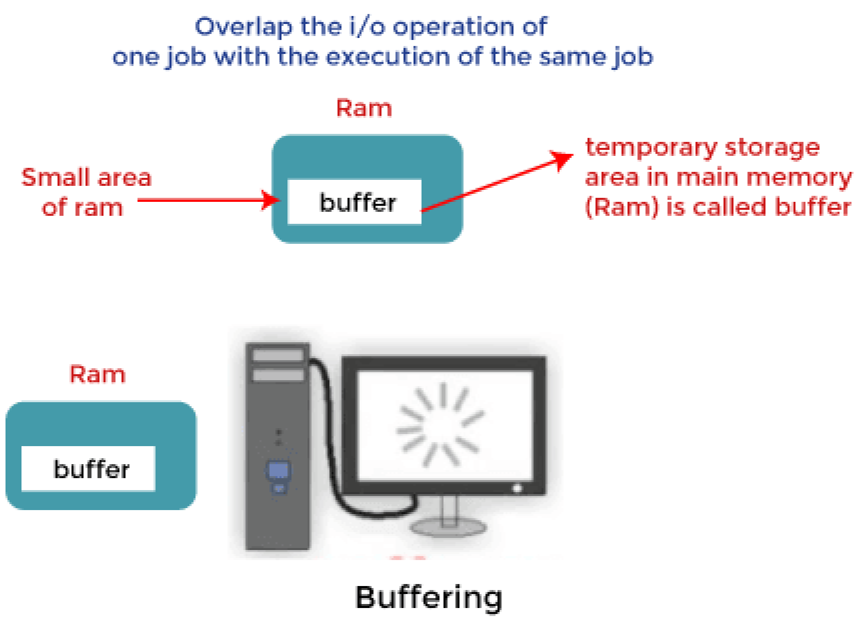
3. Circular Buffer

When more than two buffers are used, the buffers' collection is called a **circular buffer**. Each buffer is being one unit in the circular buffer. The data transfer rate will increase using the circular buffer rather than the double buffering.



How Buffering Works

In an operating system, buffer works in the following way:



Advantages of Buffer

Buffering plays a very important role in any operating system during the execution of any process or task. It has the following advantages.

- The use of buffers allows uniform disk access. It simplifies system design.
- The use of the buffer can reduce the amount of disk traffic, thereby increasing overall system throughput and decreasing response time.

Disadvantages of Buffer

- It is costly and impractical to have the buffer be the exact size required to hold the number of elements. Thus, the buffer is slightly larger most of the time, with the rest of the space being wasted.
- Buffers have a fixed size at any point in time. When the buffer is full, it must be reallocated with a larger size, and its elements must be moved. Similarly, when the number of valid elements in the buffer is significantly smaller than its size, the buffer must be reallocated with a smaller size and elements be moved to avoid too much waste.

Spooling in Operating System

SPOOL is an acronym for ***simultaneous peripheral operations online***. Generally, the spool is maintained on the computer's physical memory, buffers, or the I/O device-specific interrupts. The spool is processed in ascending order, working based on a FIFO (first-in, first-out) algorithm.

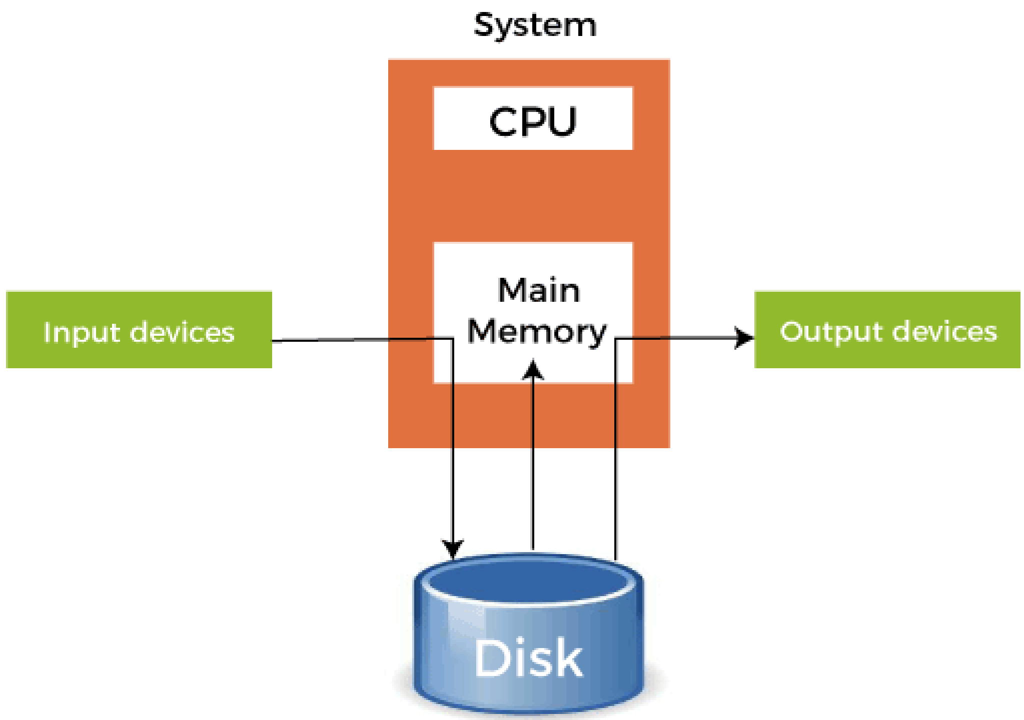
- In Operating System, we had to give the input to the CPU, and the CPU executes the instructions and finally gives the output. But there was a problem with this approach. In a normal situation, we have to deal with many processes, and we know that the time taken in the I/O operation is very large compared to the time taken by the CPU for the execution of the instructions.

How Spooling Works in Operating System

In an operating system, spooling works in the following steps, such as:

1. Spooling involves creating a buffer called SPOOL, which is used to hold off jobs and data till the device in which the SPOOL is created is ready to make use and execute that job or operate on the data.

2. When a faster device sends data to a slower device to perform some operation, it uses any secondary memory attached as a SPOOL buffer.



Example of Spooling

The biggest example of Spooling is ***printing***. The documents which are to be printed are stored in the SPOOL and then added to the queue for printing. During this time, many processes can perform their operations and use the CPU without waiting while the printer executes the printing process on the documents one-by-one.

Advantages of Spooling

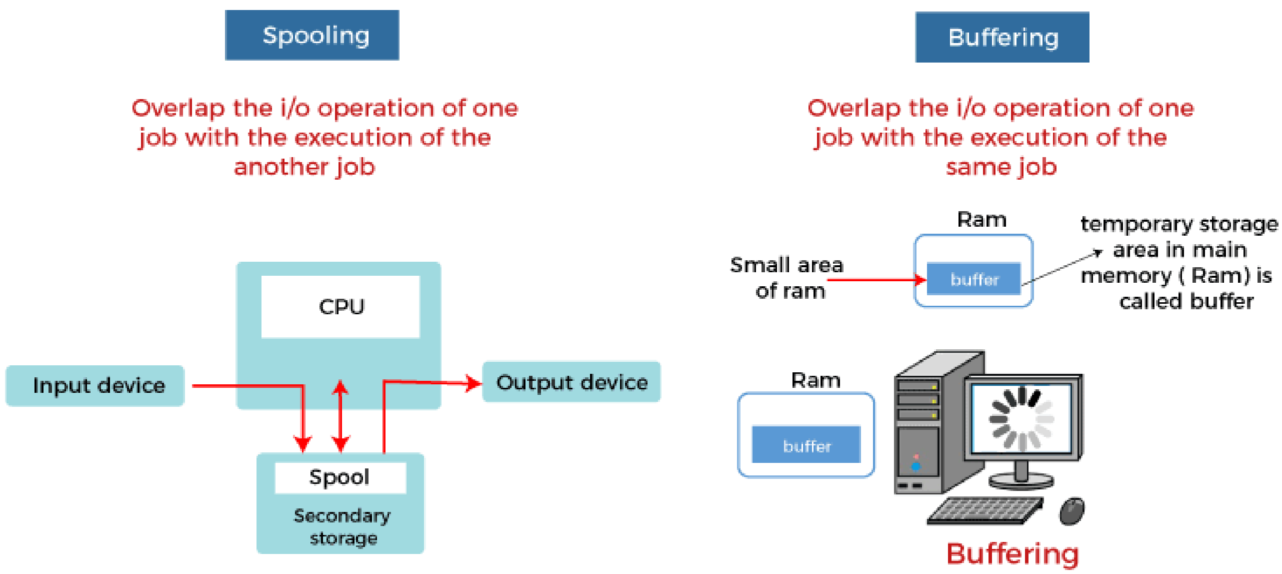
- The number of I/O devices or operations does not matter. Many I/O devices can work together simultaneously without any interference or disruption to each other.
- It allows applications to run at the speed of the CPU while operating the I/O devices at their respective full speeds.

Disadvantages of Spooling

- Spooling requires a large amount of storage depending on the number of requests made by the input and the number of input devices connected.
- Because the SPOOL is created in the secondary storage, having many input devices working simultaneously may take up a lot of space on the secondary storage and thus increase disk traffic.

Difference between Spooling and Buffering

- Spooling and buffering are the two ways by which I/O subsystems improve the performance and efficiency of the computer by using a storage space in the main memory or on the disk.

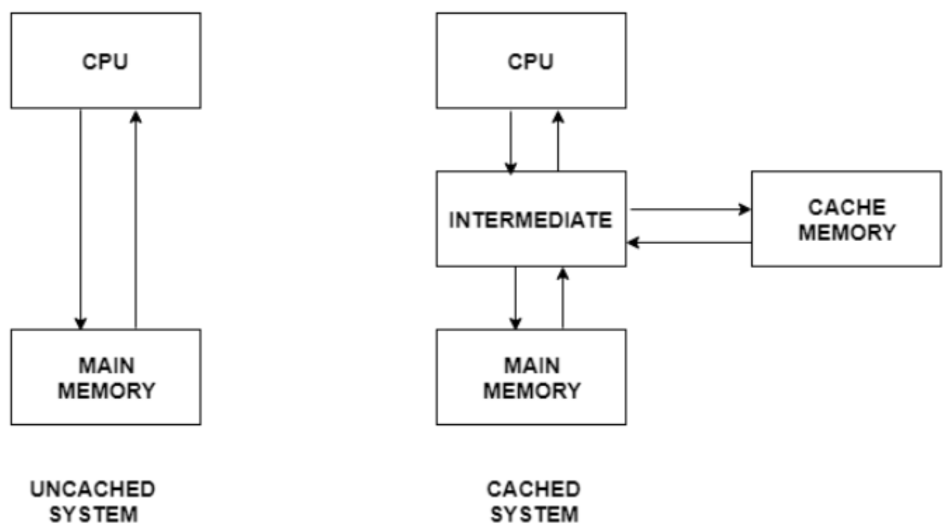


Caching in operating system

Cache is a type of memory that is used to increase the speed of data access. Normally, the data required for any process resides in the main memory. However, it is transferred to the cache memory temporarily if it is used frequently enough. The process of storing and accessing data from a cache is known as caching.

Uncached System vs Cached System

A figure to better understand the difference between cached and uncached system is as follows −



Some important points to explain the above figureare −

- In an uncached system, there is no cache memory. So, all the data required by the processor during execution is obtained from main memory. This is a comparatively time consuming process.
- In contrast to this, a cached system contains a cache memory. Any data required by the processor is searched in the cache memory first. If it is not available there then main memory is searched. The cache system yields faster results than the uncached system because cache is much faster than main memory.

Advantages of Cache Memory

Some of the advantages of cache memory are as follows −

- The memory access time is considerably less for cache memory as it is quite fast. This leads to faster execution of any process.
- The cache memory can store data temporarily as long as it is frequently required. After the use of any data has ended, it can be removed from the cache and replaced by new data from the main memory.

Disadvantages of Cache Memory

Some of the disadvantages of cache memory are as follows −

- Since the cache memory is quite fast, it is extremely useful in any computer system. However, it is also quite expensive and so is used judiciously.